Makerere University
School of Computing & IT

Course Code: CSC 2114

Course Name: Artificial Intelligence

Academic Year-2023/2024 Final Exam

Program: CSC & BSSE

Date: 8th/12/2023 Time: 4:00PM - 7:00 PM

INSTRUCTIONS:

1. Attempt ALL questions in Section A and THREE(3) in Section B
2. All questions in Section B carry equal marks.
3. Do not open this booklet until you are instructed to do so.
4. Begin each answer in Section B on a NEW page in the answer booklet
5. All rough work should be done in the answer booklet

text-speech
audio-visual
don't hear
don't see - text - sound

Section A (40 Marks)

BCd

1. Jane works with the deaf and visually impaired persons. She finds lots of challenges in communicating with them effectively. As a student of Artificial Intelligence advise Jane on how she can utilize Artificial Intelligence to improve on her communication with these people. (Hint: in your advise provide four(4) cases) [**8 marks**]

2. Human beings are known to be rational agents. Explain any four factors that influence their rationality. [**4 marks**]

3. With a set of three(3) examples, explain how the environment types affects the design of a rational agent. [**6 marks**]

4. (a) How does a tree search differ from a graph search [**2 marks**]
   (b) Using examples explain the circumstance in which you would applying the each of the following algorithms [**6 marks**]
      i. A* search
      ii. Hill climbing algorithm
      iii. MinMax algorithm

$P(A|B) = \frac{P(A \cap B)}{P(A)P(B)}$

$P(B) =$

$P(A) \cdot P(B) =$

$P(A) = \frac{P(A|B)}{P(B)}$

$P(A) \cdot P(B) = P(A|B)$

5. (a) Explain any two ways in which agents acquire knowledge. [**2 marks**]
   (b) Describe a situation(s) in which it would become handy to use each of the following knowledge representation and reasoning schemes in building a knowledge base for an Artificial Intelligence Agent. [**4 marks**]
      i. Propositional logic and First order logic
      ii. Probability

6. (a) State the product rule as used in probability. [**2 marks**]
   (b) Show how you would derive the Bayes rule from the product rule. [**4 marks**]

7. Highlight the difference(s) between Supervised learning and Unsupervised learning [**2 marks**]

Section B. Attempt any 3 questions (20 Marks each).

Question 1

(a) Using logic is one way of representing knowledge and reasoning in Artificial Intelligence. Discuss any three kinds of logic that can be used in knowledge representation and reasoning.[**9 marks**]

(b) Sarah is building a knowledge base for an automatic lighting system based /using Logic. Here are some of the sentences from Sarah's knowledge base
$S_1 = Q \wedge R \wedge S \wedge \neg W$
$S_2 = Q \wedge R$
$S_3 = Q \vee R$
$S_4 = Q$

(i) State the possible worlds for sentence S_1 [**4 marks**]

(ii) For the possible worlds stated in (bi) above. Which world(s) is/ are true for S_1 ? [**2 marks**]

(iii) Which of the sentences S_2, S_3 and S_4 entails S_1 ? [**5 marks**]

Question 2

(a) Consider the search problem in Figure 1 with start state S and goal state G. The transition costs are next to the edges, and the heuristic values are next to the states. Work out the order and path(s) each of the following search algorithms will return

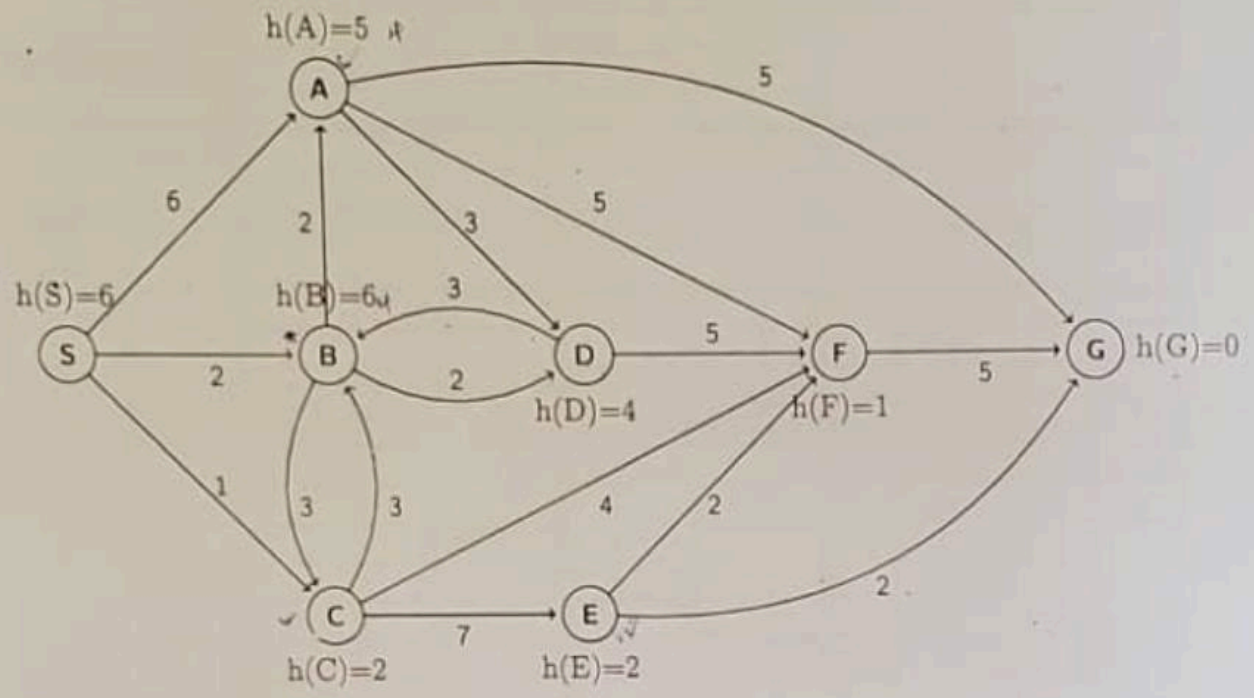

Figure 1: Search graph2

when a graph search is performed on the graph, assuming all successor functions work out in such a way that nodes are explored in alphabetical order whenever possible

(i) A^* search [**5 marks**]

(ii) Uniform cost search [**4 marks**]

(iii) Greedy Search [**3 marks**]

(b) List the paths that remain on the fringe for each of the above algorithms after reaching the goal state? [**3 marks**]

(c) Is the heuristic used at each node admissible and consistent? If it is not, which heuristic value(s) should be adjusted and to what value/ value range in order to make the heuristic admissible and consistent [**5 marks**]

Question 3

(a) State the assumption made when using the Bayes rule. **[2 marks]**

(b) A study to identify major symptoms of CoviD-19 has been carried-out by the ministry of health. A sample of the data that has been collected by the ministry is shown in Table 1. A positive indicate that patient had the symptom while a negative indicates that the patient had no symptom.

Table 1: Test Results

High temp (T)	Cough (C)	Fatigue (F)	sore throat(S)	Fever (Fe)	Body aches (B)	Covid-19 (Co)?(Yes/No)
+t	+c	+f	+s	+fe	+b	Yes
+t	+c	-f	+s	-fe	+b	Yes
+t	+c	+f	+s	+fe	-b	Yes
-t	+c	+f	-s	-fe	+b	Yes
-t	+c	-f	+s	-fe	-b	No
+t	-c	+f	+s	+fe	-b	No
-t	+c	-f	-s	-fe	+b	Yes
+t	+c	+f	+s	+fe	+b	Yes
+t	+c	-f	+s	-fe	+b	Yes
+t	+c	+f	+s	+fe	-b	Yes
-t	+c	+f	-s	-fe	+b	Yes
-t	+c	-f	+s	-fe	+b	No
+t	-c	+f	+s	+fe	+b	No
-t	+c	-f	-s	-fe	+b	Yes
+t	+c	+f	+s	+fe	+b	Yes
+t	+c	-f	+s	-fe	+b	Yes
+t	+c	+f	+s	+fe	-b	Yes
-t	+c	+f	-s	-fe	+b	Yes
-t	+c	-f	+s	-fe	-b	No
+t	-c	+f	+s	+fe	-b	No
-t	+c	-f	-s	-fe	+b	Yes
+t	+c	+f	+s	+fe	+b	Yes
+t	+c	-f	+s	-fe	+b	Yes
+t	+c	+f	+s	+fe	-b	Yes
-t	+c	+f	-s	-fe	+b	No
-t	+c	-f	+s	-fe	-b	No
+t	-c	+f	+s	+fe	-b	No
-t	+c	-f	-s	-fe	+b	Yes
-t	+c	+f	-s	-fe	+b	Yes
-t	+c	-f	+s	-fe	-b	No
+t	-c	+f	+s	+fe	-b	No

(i) Using data provided in Table 1. Calculate the following probabilities: P(T), P(C), P(F), P(Fe), P(S), P(B), P(Co), **[7 marks]**

(ii) Calculate the following probabilities: $P(T|Co)$, $P(C|Co)$, $P(F|Co)$, $P(Fe|Co)$, $P(S|Co)$, $P(B|Co)$ **[6 marks]**

(iii) Based on the results obtained in (ii) which symptom(s) is/ are most prevailing in patients with Covid-19. Explain your answer. **[2 marks]**

(iv) A patient suspected of having Covid-19 has just left the laboratory, What is the probability that patient has Covide-19 given that the patients test results were $(T = +t, C = +c, F = -f, Fe = -fe, S = -s, B = -b)$. Show your working. **[3 marks]**

Question 4

(a) What assumptions are made while formulating 'Standard' games in adversarial search? **[5 marks]**

(b) With an illustration Give a detailed description of the MiniMax algorithm. **[4 marks]**

(c) Using an illustration give a detailed description of the technique of alpha-beta (α-β) pruning algorithm. **[4 marks]**

(d) Consider the game tree in Figure 2 where A-F represent some real values. Assume the nodes are explored from left to right and standard alpha-beta (α-β) pruning is used.

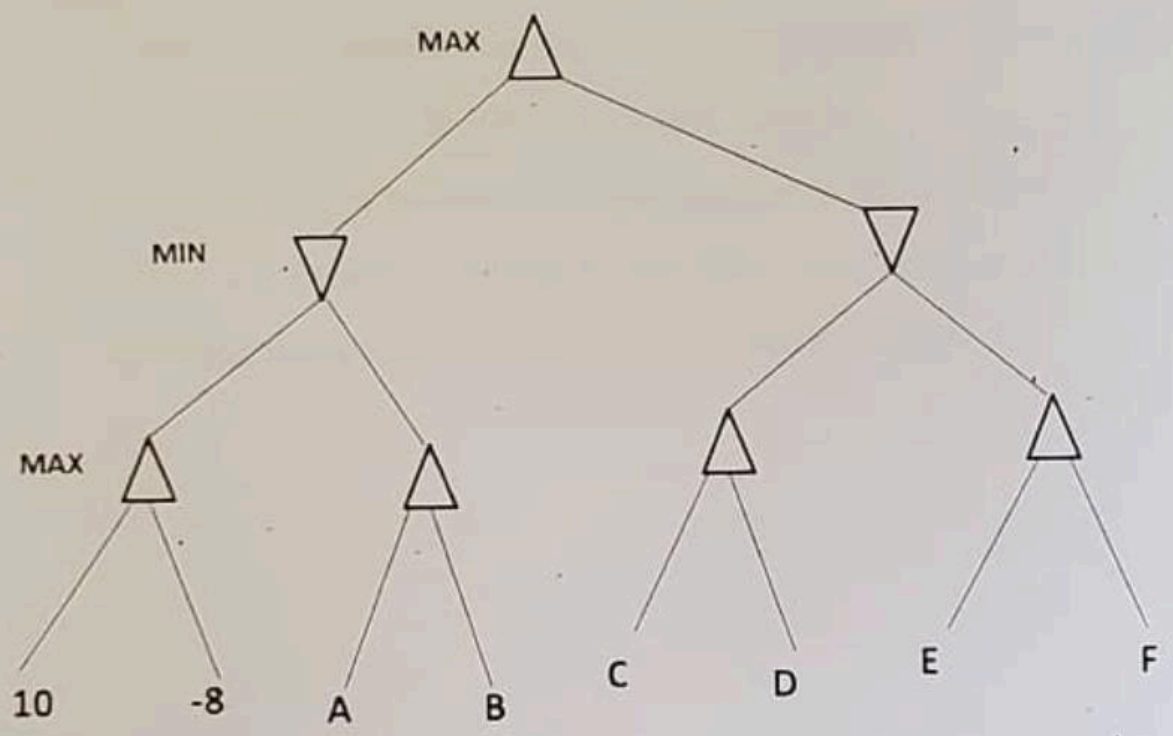

Figure 2: Game tree

(i.) Give the value(s) of **A** such that **B** is pruned. [**2 marks**]

(ii.) There are SOME values of **A** and **B** such that the subtree containing **C** and **D** is pruned. True or False . Explain why? [**2 marks**]

(iii.) Assuming that **B=6** and **A= 8**, give a value of C and D such that the subtree containing E and F is pruned.[**2 marks**]

(iv.) If you are allowed to assign *A-F* arbitrarily, what is the MAXIMUM number of leaves that can be pruned? [**1 marks**]

Question 5

(a) Discuss your understanding of machine learning [**6 marks**]

(b) Machine learning tends to be broken down into three broad classes of *learning*. Identify these classes and using examples expound on what they mean. [**6 marks**]

(c) Discuss any two machine learning classification algorithms/models you know. [**8 marks**]

End